

Sustainable hierarchical porous carbon from sugarcane bagasse for high-performance zinc-ion hybrid supercapacitors

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Highlights

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Sugarcane bagasse used to synthesize sustainable porous carbon material.

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MgO templating enhances porosity and graphitic order in carbon structure.

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Modified carbon achieves 482 F/g capacitance and 268 mAh/g capacity at 1 A/g.

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ZIHSC shows 80 % capacity retention after 1000 cycles with high efficiency.

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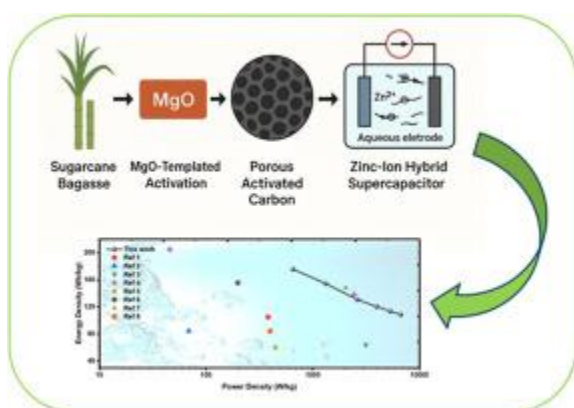
Outperforms prior systems, promoting biomass use in energy storage devices.

Abstract

The growing demand for sustainable and high-performance energy storage systems has driven interest in bio-derived carbon materials, particularly for use in zinc ion hybrid supercapacitors (ZIHSCs). In this study, sugarcane bagasse was utilized as a low-cost and renewable precursor for the synthesis of activated carbon, with its structure significantly

enhanced through a templating method using magnesium oxide. The resulting modified sugarcane bagasse-derived activated carbon (SAC) exhibited a highly porous and more ordered carbon framework. These structural improvements enabled superior ionic diffusion and electrical conductivity, making the material highly effective as a ZIHSC cathode. The modified SAC-based ZIHSC system achieved an impressive specific capacitance of 482 F/g and a specific capacity of 268 mAh/g at a current density of 1 A/g. Moreover, it demonstrated excellent cycling stability with 80 % capacity retention over 1000 cycles and nearly ideal charge transfer and reversibility. Compared with previously reported ZIHSC technologies, this system offers a compelling alternative for next-generation energy storage solutions, highlighting the potential of agricultural waste valorization in sustainable device development.

Graphical abstract



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Keywords

Sugarcane bagasse

Activated carbon

Zinc ion hybrid supercapacitor

Templated porous carbon

Sustainable energy storage

1. Introduction

As global energy consumption continues to escalate concurrently with mounting environmental concerns, the imperative for dependable and renewable energy storage solutions has grown increasingly critical. Achieving efficient energy storage transcends mere technical ambition; rather, it constitutes an essential component in the pursuit of environmental sustainability and the broader objective of establishing a resilient and sustainable future [1]. High-performance energy storage systems (ESSs), renowned for their long cycle life and large capacity, are gaining significance as electric vehicles and portable electronic devices become increasingly prevalent in everyday life. Batteries, as one of the most widely used energy storage systems, typically provide a high energy density ranging from 200 to 300 Wh/kg [2,3], but often face limitations such as low power density (100–1000 W/kg) and a relatively short lifespan, typically lasting only 400 to 1200 charge-discharge cycles [4,5]. In comparison, supercapacitors offer significantly higher power density and much longer cycle life [6,7], although their energy density remains comparatively low [2,8]. To address these limitations, researchers have explored integrating battery-type and capacitive-type materials into a single device, commonly referred to as a hybrid supercapacitor or supercapattery, to harness the strengths of both systems [9].

These hybrid supercapacitors successfully combine the redox reaction mechanisms typical of battery electrodes with the fast adsorption/desorption processes characteristic of supercapacitor electrodes, thereby achieving a balanced and enhanced electrochemical performance [[10], [11], [12]]. Zinc-ion hybrid supercapacitors (ZIHSCs) are emerging energy storage devices that merge the advantages of zinc anodes with those of supercapacitors, resulting in enhanced electrochemical performance. These systems have attracted considerable scientific attention due to their affordability, improved safety, and impressive energy and power densities [13]. With a theoretical capacity of around 820 mAh g⁻¹, low risk associated with the Zn metal anode, compatibility with aqueous electrolytes, and a standard reduction potential of -0.76 V (vs. Zn²⁺/Zn), ZIHSCs present a promising alternative in the field of next-generation energy storage [[13], [14], [15], [16], [17], [18]].

Carbon-based materials, particularly activated carbon, are widely used as cathode materials in ZIHSCs due to their excellent electrical conductivity, chemical stability, large surface area, and tunable pore structures [[19], [20], [21], [22], [23]]. Activated carbon cathode materials derived from biomass have exhibited remarkable surface areas, reaching up to 3000 m²/g, and specific capacitance values typically ranging between 200 and 300 F/g [13,24]. The use of biomass is particularly appealing for ZIHSC applications due to its low cost, ease of processing, and environmentally friendly nature [13,14]. However, maintaining stability between carbon-based electrode materials and electrolytes remains a significant challenge, primarily attributed to the mismatch in pore size

distribution of the carbon materials. The ion adsorption/desorption dynamics at the carbon cathode do not synchronize well with the Zn^{2+} deposition/stripping behaviour at the zinc anode, resulting in significant cathodic hysteresis. Consequently, the overall capacity of the system remains well below the theoretical potential of the zinc anode [10]. Therefore, tailoring the structural features and surface chemistry of carbon materials is crucial for enhancing the capacitive behaviour and overall performance of ZIHSCs [10,25].

Researchers have made considerable progress in advancing ZIHSCs, particularly through the use of biomass-derived carbon as cathode material. In our previous work, activated carbon derived from jute sticks was utilized as the cathode, demonstrating stable and reliable electrochemical performance for energy storage applications [26,27]. In another study, we employed activated carbon derived from basil plants, which also showed promising results when directly applied as the cathode in ZIHSCs [28]. Similarly, Hongxia and colleagues reported the use of hierarchical porous carbon derived from olive leaves as an effective cathode material for anti-self-discharge ZIHSCs [29]. Additionally, Burak and co-workers introduced activated carbon obtained from hemp biomass as a viable cathode option [19]. More recently, sugarcane bagasse-derived activated carbon has been proposed as an efficient cathode material for ZIHSCs, attributed to its naturally occurring microfibrinous structure [30,31].

Beyond its favorable morphological characteristics, sugarcane was chosen as the carbon source in this study due to its widespread use in prior research on carbon-based materials. Sugarcane bagasse, in particular, stands out as a highly promising carbon precursor because of its rich composition, approximately 20 % lignin, 45 % cellulose, and 25 % hemicellulose [32]. Its abundant availability further enhances its appeal, accounting for nearly 90 % of the by-products from processed sugarcane. In Indonesia, sugarcane production currently reaches around 350 million tons, with an average productivity of 66.9 tons per hectare. For every ton of milled sugarcane, about 180–280 kg of bagasse residue is generated [33].

This study focuses on enhancing the performance of ZIHSCs by utilizing activated carbon derived from sugarcane bagasse. Conventional activation methods often yield predominantly microporous structures, which hinder electrolyte ion movement and reduce overall efficiency. To overcome this limitation, magnesium oxide (MgO) is employed as a templating agent because it can be easily removed after carbonization and effectively promotes the formation of mesoporous structures. This results in a more open pore network that facilitates faster ion transport and improved electrochemical accessibility. The mesoporous activated carbon obtained in this way demonstrates superior electrochemical performance as a cathode material for ZIHSCs compared to unmodified

carbon, confirming the effectiveness of this modification strategy. Furthermore, this research highlights sugarcane bagasse as a cost-effective and renewable carbon source, reinforcing its potential in the development of sustainable and high-performance energy storage systems.

2. Experimental

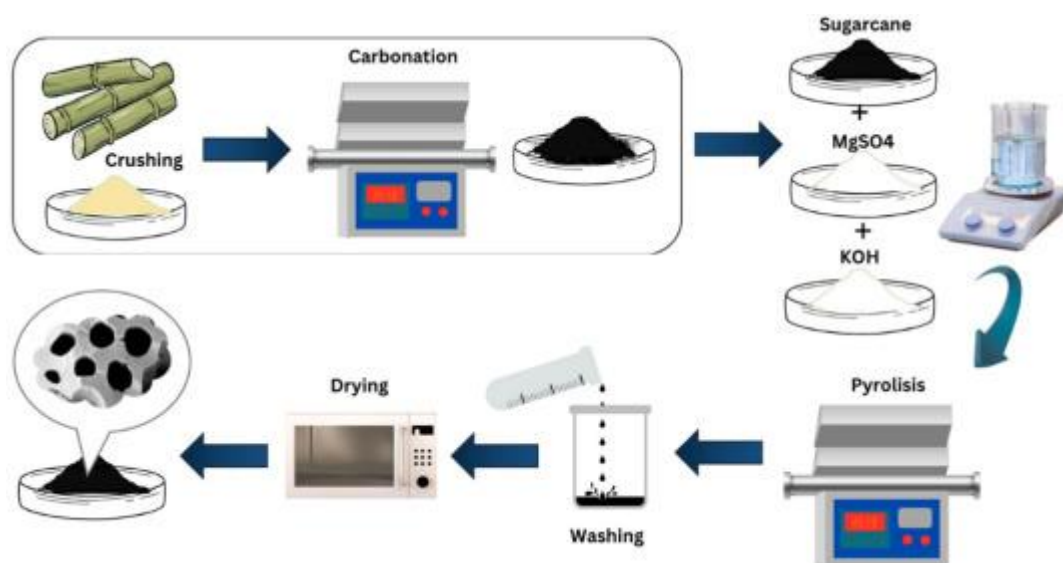
2.1. Materials

The materials used in this study were sourced from commercial suppliers. Zinc sulfate heptahydrate ($\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$) was obtained from Loba Chemie, while zinc foil (0.25 mm thickness) and desiccated sugarcane bagasse were collected from Java Island, Indonesia. Additional chemicals included potassium hydroxide (KOH), magnesium sulfate (MgSO_4), 1-methyl-2-pyrrolidone (NMP), polyvinylidene fluoride (PVDF, HSV900), and Super P conductive carbon, all procured from Sigma-Aldrich or MSE Supplies. Whatman glass fiber paper was used as a separator, and AL KIG, Dammam, supplied high-purity nitrogen gas (99.999 %).

2.2. Methods

Sugarcane bagasse was initially sourced from local markets and farms on the island of Java, Indonesia. The raw material was thoroughly rinsed with deionized (DI) water to eliminate surface impurities and contaminants. After washing, the material was dried at 120 °C for 2 h in a hot air oven, ground into fine particles using a mechanical grinder, and then sieved through a 300 μm mesh to obtain a uniform particle size. The dried bagasse was further pulverized using a mortar and pestle. For carbonization, the powdered material was placed in a muffle furnace and heated at 300 °C for 2 h. The carbonized product, referred to as sugarcane bagasse carbon (SC), was then used for further processing.

To enhance porosity, a modified method based on literature [34,35] was applied. SC was mixed with potassium hydroxide (KOH) and magnesium sulfate (MgSO_4) in a weight ratio of 1:3:2, along with 20 mL of DI water. This mixture was left to dry overnight. The activation process was carried out at 700 °C for 2 h under a nitrogen atmosphere. The resulting carbon was sequentially washed, first with 2 M sulfuric acid (H_2SO_4) to remove the MgO template, and then with deionized water until the filtrate reached a neutral pH. The final product was filtered, dried at 100 °C for 3 h, and collected as modified sugarcane bagasse-derived activated carbon (SAC). [Fig. 1](#) illustrates the schematic diagram of the preparation steps. For comparison, SAC was also prepared using the same procedure but without the addition of MgSO_4 , thereby excluding the MgO templating effect. Theoretically, this process yields about 10 % of carbon product relative to the initial mass of the precursor.



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Fig. 1. Schematic representation of the preparation process of modified SAC.

2.3. Fabrication of ZIHSCs

The cathodes for the ZIHSCs were prepared by mixing 70 % of active material (SC, SAC, or modified SAC), 20 % Super P conductive carbon, and 10 % PVDF binder in 1-methyl-2-pyrrolidone (NMP). The mixture was stirred thoroughly for 30 min to form a uniform and stable slurry. This viscous black slurry was then carefully drop-cast onto circular graphite foil electrodes. The coated electrodes were vacuum-dried at 80 °C to remove any residual solvent. For cell assembly, coin-type ZIHSCs were constructed using the prepared cathode, a zinc foil anode, and a glass fiber separator, with 2 M zinc sulfate solution serving as the electrolyte.

2.4. Materials characterization

The structure and surface morphology of the modified activated carbon derived from sugarcane bagasse were analyzed using a range of characterization techniques. X-ray diffraction (XRD) was performed with a Rigaku Miniflex-II diffractometer, utilizing Cu K α radiation ($\lambda = 0.15416$ nm), operated at 30 mA current and 40 kV voltage. Field Emission Scanning Electron Microscopy (FESEM), combined with Energy Dispersive Spectroscopy (EDS), along with High-Resolution Transmission Electron Microscopy (HR-TEM), were used to observe the surface features and analyze the elemental composition of the carbon material. Raman spectroscopy was employed to investigate the vibrational characteristics of the carbon using an iHR320 HORIBA Raman spectrometer, operated with a 300 mW,

532 nm green laser and a CCD detector. Additionally, nitrogen adsorption–desorption isotherms and pore size distribution were measured using the Brunauer–Emmett–Teller (BET) method on a Micromeritics ASAP 2020 surface area and porosity analyzer.

2.5. Electrochemical measurements and analysis

The electrochemical performance of the ZIHSCs, utilizing the synthesized activated carbon as the cathode material, was evaluated using various techniques with two different potentiostat systems: the CHI 760 and the LAND IT battery testing system. The CHI 760 facilitated measurements including cyclic voltammetry (CV), galvanostatic charge-discharge (GCD), and electrochemical impedance spectroscopy (EIS). Meanwhile, the LAND IT system was specifically employed to evaluate the cycling stability and rate capability of the devices.

In addition, following the comprehensive electrochemical measurements, key performance parameters including specific capacitance (C in F/g), energy density (E in Wh/kg), and power density (P in W/kg) were calculated using the following equations (1), (2), (3) [36,37].(1)(2)(3)

In these equations, V represents the voltage profile during the discharge phase as a function of time (in volts), ΔV denotes the operating voltage window of the ZIHSCs (in volts), Δt is the discharge time (in seconds), and i refers to the current density (in amperes per gram).

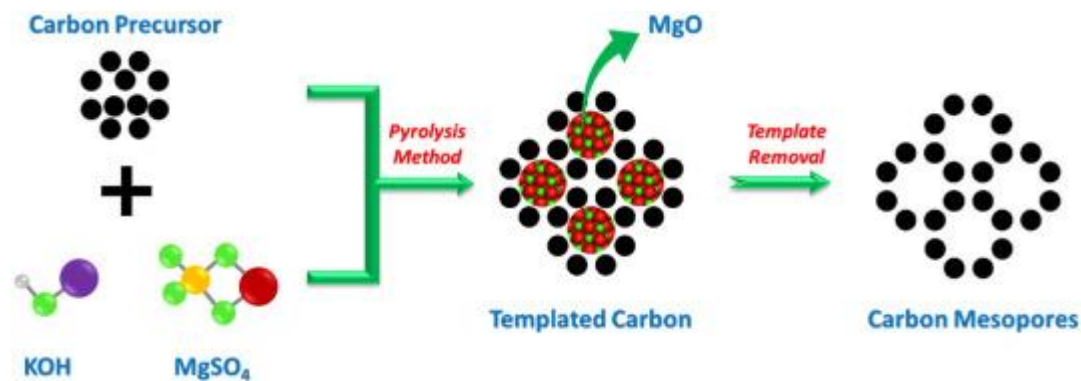
3. Results and discussion

To enhance the porosity of the carbon structure derived from sugarcane bagasse, a templating method was employed. In this work, MgO was chosen as the template material due to its excellent thermal stability and its ability to form in situ from a MgSO_4 precursor at relatively low temperatures (less than 1000 °C) in the presence of carbon [38]. Moreover, MgO can be effectively removed through acid washing, which facilitates the formation of well-developed pores as the template is dissolved and extracted from the carbon matrix [39].

To further explain the mechanism of the templating process, carbon derived from sugarcane was initially mixed with magnesium sulfate (MgSO_4), which acted as the precursor for the MgO template, and potassium hydroxide (KOH), which served as the chemical activating agent. This combination likely enabled uniform dispersion and intimate contact among the components, allowing for coordinated interaction during the pyrolysis process. Consequently, the formation of the MgO template and the generation of porosity through activation could occur simultaneously and in close proximity. The formation of

MgO in this context can be represented by the following reaction in equation (4) [38]. $(4)\text{MgSO}_4 \rightarrow \text{MgO(s)} + \text{SO}_2\text{(g)} + \frac{1}{2} \text{O}_2\text{(g)}$

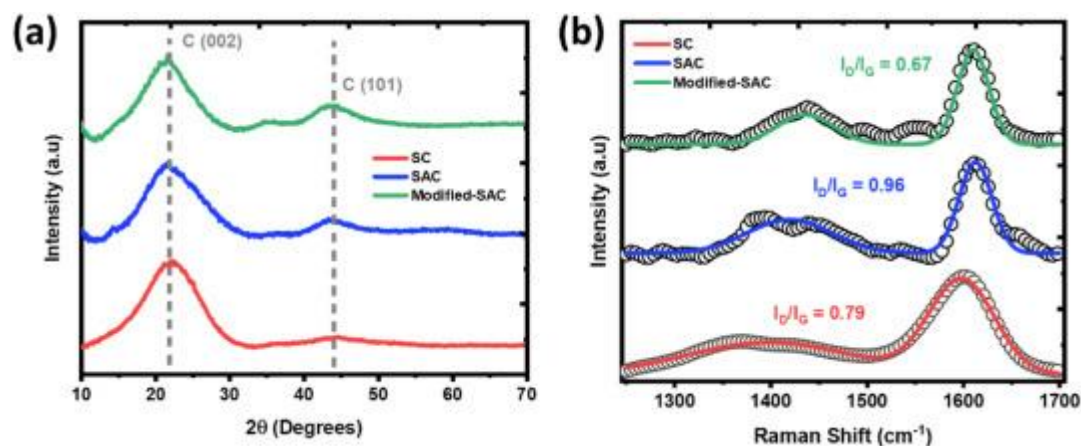
Since the formation of MgO occurs at the same sites as the development of porosity in the carbon matrix, the size of the resulting pores can be influenced by the MgO particles. This process can be visualized as MgO particles being surrounded by carbon, as illustrated schematically in Fig. 2. Subsequently, acid washing dissolves the MgO particles, leaving behind a porous structure within the carbon morphology.



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Fig. 2. Schematic illustration of porosity enhancement via the templating approach.

To investigate the formation and intrinsic characteristics of the synthesized carbonaceous materials, namely raw SC, SAC, and modified SAC, a comprehensive suite of characterization techniques was employed. The XRD analysis revealed similar diffraction patterns for SC, SAC, and modified SAC, as shown in Fig. 3(a). All samples displayed two broad peaks centred at $2\theta \approx 22^\circ$ and $2\theta \approx 44^\circ$, corresponding to the (002) and (101) planes of graphite, respectively [40,41]. These peaks indicate the presence of both semi-graphitic and amorphous carbon structures. The absence of additional diffraction peaks confirms the successful formation of carbon and the effective removal of the templating material. Interestingly, variations were observed in the full width at half maximum (FWHM) of the (002) peak across the samples. Generally, a broader XRD peak or larger FWHM is indicative of a more amorphous structure, while a sharper peak reflects a higher degree of graphitic ordering [42].



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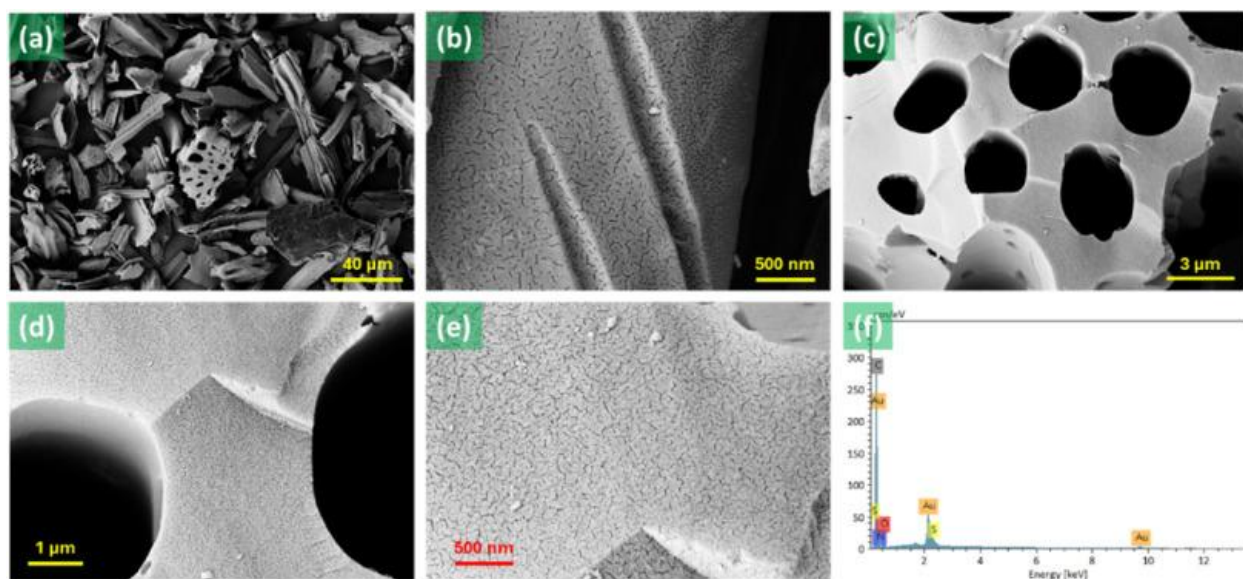
Fig. 3. (a) XRD patterns and (b) Raman spectra of SC, SAC, and modified SAC samples.

The FWHM values at $2\theta \approx 22^\circ$ for SC, SAC, and modified SAC were found to be approximately 8.0° , 8.5° , and 6.3° , respectively. The SAC exhibited the broadest peak, which can be attributed to the chemical activation process that introduced significant porosity, resulting in a more disordered structure with an increased surface area. In contrast, the modified SAC showed a narrower peak, indicating a higher degree of structural ordering. This enhancement in order is likely due to the interaction between MgO and carbon at the interface during pyrolysis, which may facilitate local alignment of carbon atoms and the formation of graphitic domains [43]. The improved ordering in the modified SAC is further supported by the more pronounced intensity of the peak near $2\theta \approx 44^\circ$. These findings indicate that the modified SAC not only possesses a high degree of porosity resulting from the activation process, which contributes to an increased surface area, but also demonstrates local graphitization, which could potentially enhance the electrical conductivity of the carbon material.

Furthermore, Raman spectroscopy serves as a complementary technique to XRD for investigating the degree of order and disorder in carbon structures. In this study, Raman spectroscopy was performed on all carbon samples, and the results are presented in Fig. 3(b). The degree of structural ordering in the carbon materials can be evaluated by examining the D band and G band [44], typically located around 1400 cm^{-1} and 1600 cm^{-1} , respectively. In the Raman spectra, the D-band position shows slight variations among the three carbon samples. These shifts arise from differences in defect concentration, structural disorder, and local strain effects induced during activation and MgO templating. Moreover, the ratio of the intensities of these bands (I_D/I_G), calculated based on the area under each peak, is commonly used to assess the structural disorder. A higher I_D/I_G ratio

indicates a greater degree of disorder, corresponding to a more amorphous carbon structure, while a lower ratio suggests a more ordered or graphitized structure [44]. The I_D/I_G values for SC, SAC, and modified SAC were approximately 0.67, 0.96, and 0.79, respectively. The highest value observed for SAC is attributed to the activation process, which introduces significant porosity and increases structural disorder, consistent with the findings from the XRD analysis. In contrast, the reduced I_D/I_G value for the modified SAC indicates the presence of more ordered structures, likely due to the formation of local graphitic domains as discussed in the XRD results. Therefore, both XRD and Raman analyses confirm that the modified SAC exhibits an enhanced structure characterized by a combination of high porosity and increased graphitic ordering. This dual structural improvement is advantageous for applications such as electrodes in ZIHSCs, where both high ion diffusivity and good electrical conductivity are critical [45].

To further investigate the formation of porous structures in the prepared carbon materials from a visual perspective, FESEM analysis was performed. FESEM imaging was conducted for SC, SAC, and modified SAC samples. The FESEM results for the modified SAC are shown in Fig. 4, while the corresponding images for SC and SAC are provided in Fig. S1. At low magnification, all carbon samples displayed a microfibrinous structure with a wood-like surface, as shown in Fig. 4(a), Figure S1(a), and Fig. S1(d). This structure reflects the natural morphology of sugarcane bagasse. When observed at moderate magnification, hole structures began to appear in the SAC and modified SAC samples (Fig. 4(c, d) and Fig. S1(e)), whereas such features were absent in SC (Fig. S1(b)). This suggests that the activation process plays a key role in creating surface porosity, which is not observed in the SC sample due to the lack of chemical activation, as it was only subjected to carbonization.



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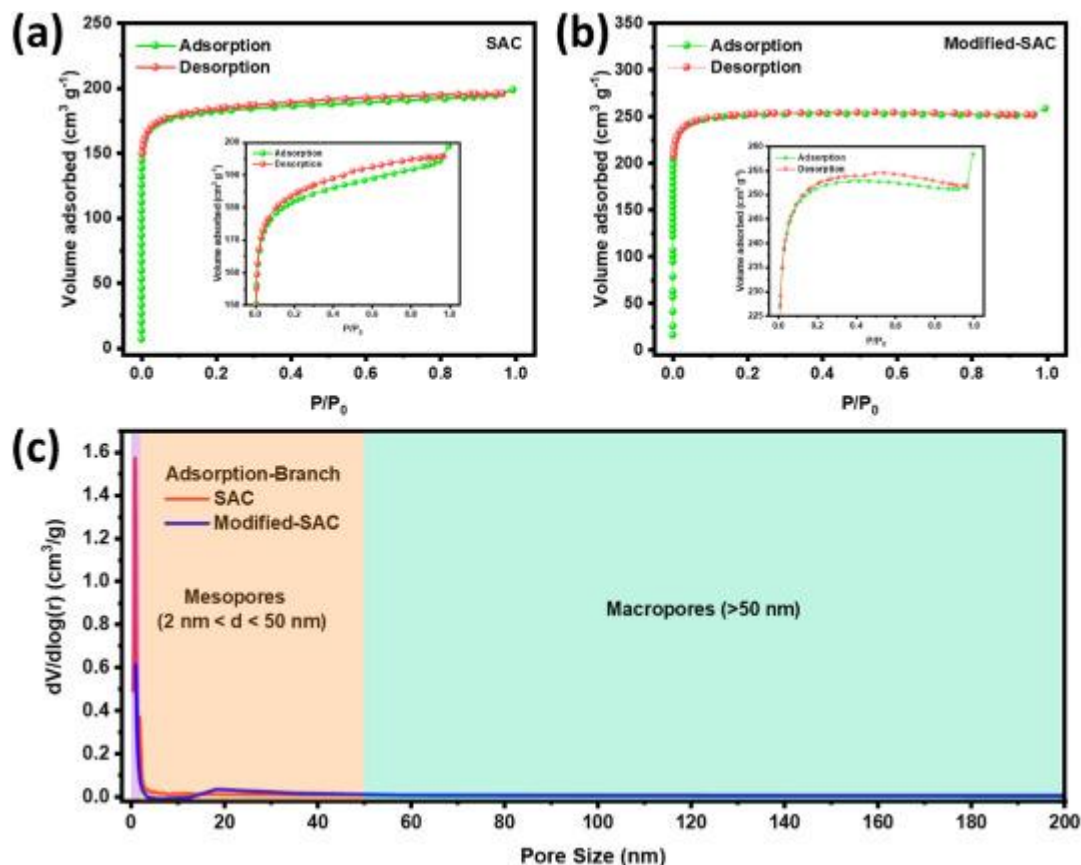
Fig. 4. (a–e) FESEM images at different magnifications and (f) EDS spectrum of modified SAC.

Notably, a large number of open porous structures were observed in the modified SAC, a smaller amount in SAC, and none in SC, as illustrated in [Fig. 4\(b–e\)](#), [Fig. S1\(c\)](#), and [Fig. S1\(f\)](#), respectively. These observations clearly confirm the successful templating effect of MgO, which facilitates the formation of a more open and accessible porous network within the activated carbon. This finding supports the earlier discussion that the modified SAC maintains a highly porous structure, despite the appearance of local graphitization resulting from interactions with the MgO templating material. Furthermore, this combination of enhanced porosity and partial graphitization is particularly advantageous for energy storage applications, such as supercapacitors, where efficient ion diffusion and improved electrical conductivity are essential for optimal performance.

Moreover, elemental analysis of the modified SAC was performed using EDS spectroscopy to assess the purity of the resulting carbon material. The EDS spectrum is shown in [Fig. 4\(f\)](#). As observed, the most prominent peak corresponds to carbon, indicating that carbon is the dominant element present in the modified SAC sample. The presence of gold (Au) is attributed to the gold coating applied to the sample prior to EDS analysis, which improves conductivity. The nitrogen signal is likely due to the natural composition of sugarcane, which may contain nitrogenous compounds [46]. The detected sulfur may result from residual sulfur species originating from either magnesium sulfate (MgSO_4) used in the templating process or sulfuric acid (H_2SO_4) used during the washing step. Nonetheless, the significantly higher carbon content compared to other elements confirms the successful formation of carbon in the modified SAC.

Furthermore, to quantitatively evaluate the porosity of SAC and modified SAC, BET analysis was conducted, as presented in [Fig. 5](#). SAC and modified SAC were selected for this analysis because both underwent activation or modification processes designed to enhance porosity, whereas raw sugarcane carbon was excluded since it was neither activated nor modified and therefore does not provide a meaningful comparison. The nitrogen adsorption–desorption isotherms of SAC ([Fig. 5\(a\)](#)) and modified SAC ([Fig. 5\(b\)](#)) reveal important information about surface area and pore structure. The BET surface areas were determined to be $564 \text{ m}^2 \text{ g}^{-1}$ for SAC and $762 \text{ m}^2 \text{ g}^{-1}$ for modified SAC. This increase in surface area following MgO modification demonstrates the development of a more open

and accessible pore structure, which is consistent with the morphological observations from SEM.



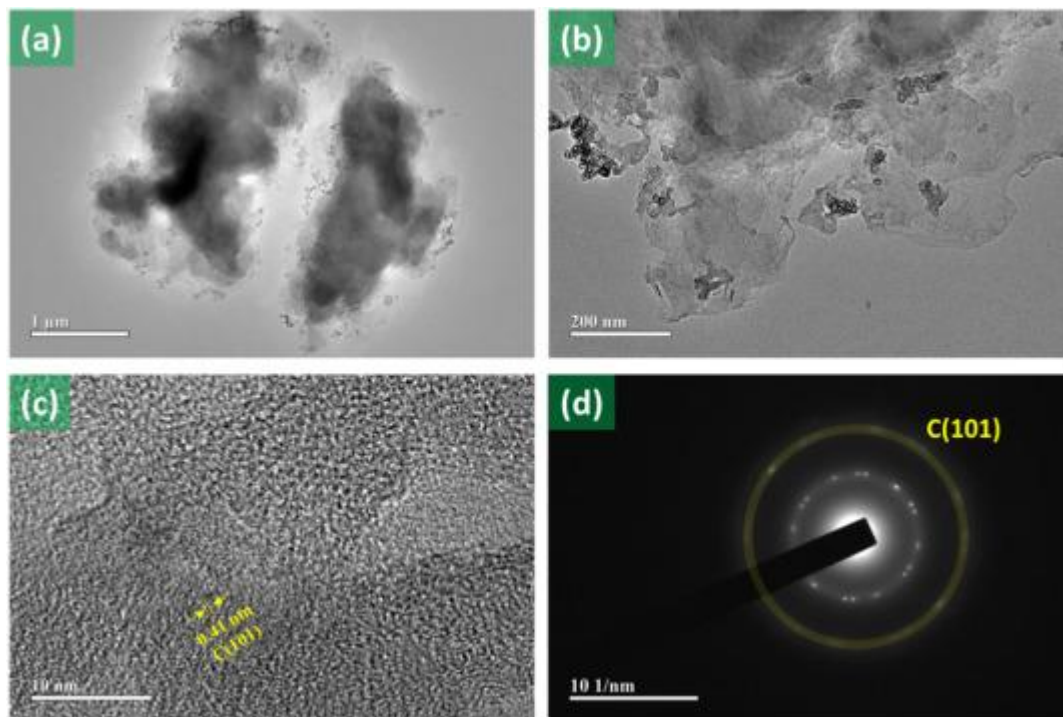
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Fig. 5. (a) N₂ adsorption–desorption isotherms of SAC, (b) N₂ adsorption–desorption isotherms of modified SAC, and (c) pore size distribution of both carbons.

The enhanced porosity of the modified SAC is further confirmed by the pore size distribution profiles obtained using the Barrett–Joyner–Halenda (BJH) method (Fig. 5(c)). SAC exhibits a pore distribution composed of 71 % micropores, 24 % mesopores, and 5 % macropores, as calculated by integration of the pore distribution curve. After MgO modification, the distribution shifts markedly to 32 % micropores, 62 % mesopores, and 6 % macropores. This transformation indicates that part of the microporous structure evolves into mesopores due to the templating effect of MgO. Such a shift is advantageous because an excessive fraction of micropores, although beneficial for high surface area, can hinder ion transport and limit rate performance, leading to diminished capacity at high current densities. By contrast, the significant increase in mesoporosity provides a more

open and interconnected pore network, which facilitates faster ion diffusion and better accessibility to active sites. These structural improvements are therefore expected to directly contribute to the enhanced electrochemical performance of electrodes based on the modified SAC material.

Moreover, given that earlier results suggested the presence of local graphitization in the modified SAC, in addition to its improved porosity, a more detailed examination was necessary. To specifically observe the graphitic features within the carbon morphology, HRTEM analysis was performed. Fig. 6 displays the HRTEM images of the modified SAC at both low and high magnifications, along with its selected area electron diffraction (SAED) pattern. Interestingly, at low and moderate magnification, nanosheet-like structures were observed on the surface of the particles, as shown in Fig. 6(a and b). This feature may result from morphological cracking caused by the dissolution of the material in ethanol during sample preparation. The cracking is likely associated with the presence of abundant mesopores within the carbon structure. The combination of mesoporous characteristics, along with the macroporous and microporous features previously observed in SEM images and BET analysis, may contribute to the formation of these nanosheet-like structures attached to the particle surface.



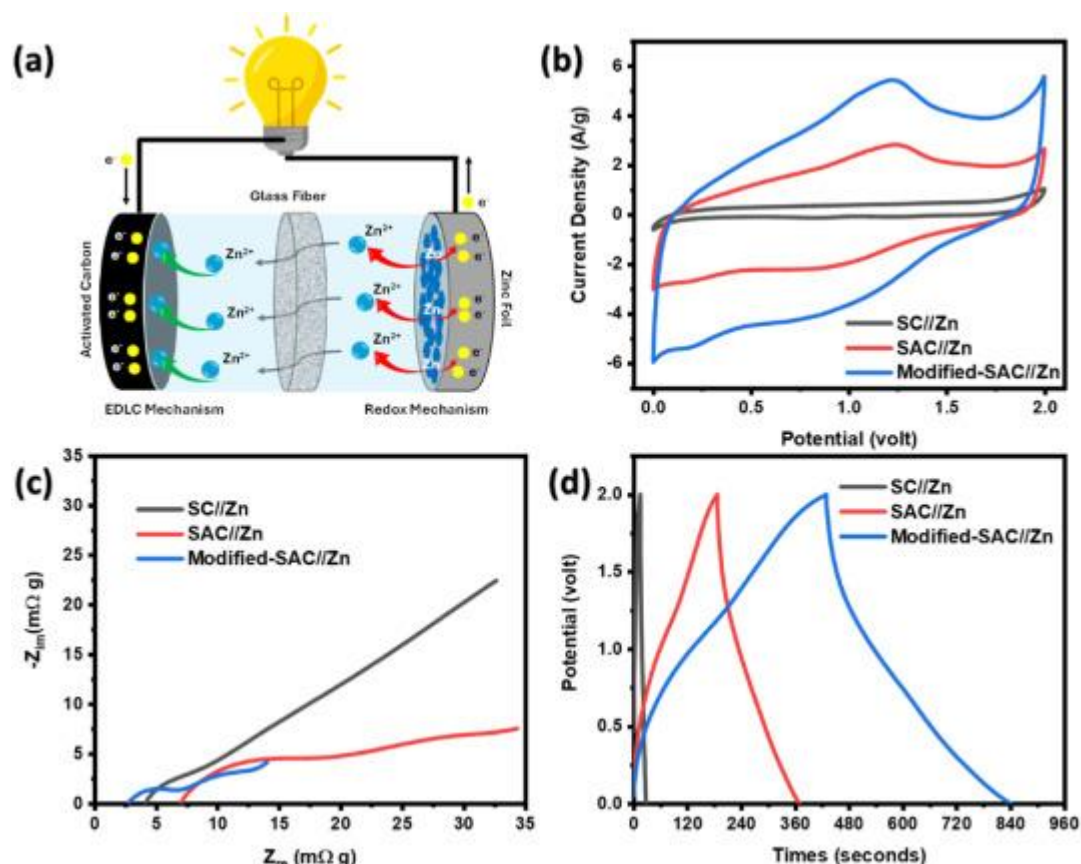
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Fig. 6. (a–c) HRTEM images at different magnifications and (d) SAED pattern of modified SAC.

Furthermore, within this morphological structure, regions of local graphitization were observed, supporting the earlier findings from XRD and Raman spectroscopy, which indicated enhanced graphitic character in the modified SAC. Specifically, a d-spacing of approximately 0.41 nm was measured in the structure of the modified SAC, as shown in [Fig. 6\(c\)](#). This spacing corresponds to the (101) lattice plane, consistent with the diffraction peak identified in the XRD analysis. Additionally, the selected area electron diffraction (SAED) pattern reinforces this observation by displaying circular diffraction rings associated with the (101) plane, as presented in [Fig. 6\(d\)](#).

These findings further confirm that the modified SAC possesses not only a highly porous structure but also an increased degree of graphitization. These structural characteristics play a crucial role in influencing the electrochemical performance of electrodes utilizing modified SAC as the active material. In this study, the modified SAC was specifically employed as the cathode material for a ZIHSC.

As briefly mentioned in the previous discussion, the modified SAC was utilized as the cathode material for a ZIHSCs. For comparison, other prepared carbon materials, including SAC and SC, were also evaluated. The schematic diagram of the ZIHSCs configuration using activated carbon as the cathode material is shown in [Fig. 7\(a\)](#). In this setup, activated carbon serves as the cathode, zinc foil is used as the anode, glass fibre acts as the separator, and ZnSO_4 is used as the electrolyte. Conceptually, the ZIHSCs operate through different charge storage mechanisms at each electrode. On the cathode side, the carbon electrode primarily follows the electrical double layer capacitance (EDLC) mechanism. In contrast, the zinc foil anode undergoes a reversible redox reaction involving the stripping and plating of zinc ions [\[47\]](#), which the following reversible redox reaction can represent as in equation [\(5\)](#).(5)



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Fig. 7. (a) Schematic of ZIHSC system. (b) CV curves at scan rate of 10 mV/s; (c) Nyquist plots; (d) GCD curves at current density of 2 A/g for ZIHSCs with different cathode materials.

Since each electrode operates through a different charge storage mechanism, it is essential to maintain charge balance within the ZIHSCs full cell system. Achieving this balance typically influences the shape of the CV curve, often resulting in a unique profile for the ZIHSCs full cell. An optimized charge balance between the cathode and anode generally leads to a larger CV curve area, which corresponds to improved performance in terms of specific capacity and energy density [48].

The CV curves at a scan rate of 10 mV/s for the full-cell ZIHSC systems using different cathode materials—SC, SAC, and modified SAC—are shown in Fig. 7(b). Both the SAC//Zn and modified SAC//Zn systems exhibit distinct CV profiles, characterized by a larger EDLC-type curve combined with well-defined redox peaks around 1.2 V. This unique feature arises from the charge balance between the EDLC process at the carbon cathode and the

redox reaction of the zinc anode. The appearance of the peak near 1.2 V is further supported by our simulation results ([Fig. S2](#)), where separate CVs of AC and zinc were modeled. The zinc electrode shows a redox peak near its standard potential of -0.78 V vs. SHE, and when combined with the capacitive response of AC, the simulated full cell displays an anodic peak around 1.2 V. This simulated CV, provided in [Supporting Information 2](#) and based on our previous work [48], aligns well with the experimental CV results, confirming that the observed peak originates from the zinc anode. In contrast, the SC//Zn system does not exhibit this combined behavior, likely because the EDLC contribution from the SC cathode interacts only with the non-faradaic region of the zinc anode, thereby preventing the appearance of redox peaks in its CV profile.

Furthermore, the CV curve of the modified SAC//Zn system displayed the largest enclosed area among all the tested full cell configurations. A larger CV curve area generally indicates a greater number of active sites participating in the energy storage process [49]. This suggests that the modified SAC//Zn system possesses superior specific capacity and energy density compared to the other systems evaluated. This is primarily attributed to the enhanced porosity and structural features of the modified SAC cathode material, which facilitate a more effective interaction between the ions and the electrode's active surface area.

EIS, typically represented by a Nyquist plot, is a valuable technique for investigating the internal electrochemical behavior of electrode materials during the energy storage process, complementing CV analysis. EIS is particularly effective in evaluating charge transfer resistance, which reflects the ease or difficulty of ion and electron interactions at the electrode–electrolyte interface. This resistance is commonly determined by the diameter of the semicircular region in the Nyquist plot [41]. In addition to resistance analysis, the Nyquist plot also provides insight into the dominant charge storage mechanism within the electrode. This is typically assessed in the low-frequency region, where the plot displays a linear segment. The slope of this line reveals the nature of the charge storage: a lower slope indicates a diffusion-controlled mechanism, while a steeper slope suggests a capacitive behaviour.

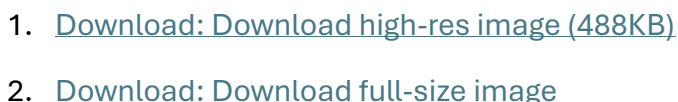
Interestingly, as illustrated in [Fig. 7\(c\)](#), the Nyquist plots of ZIHSCs with different cathode materials exhibit typical features of an energy storage system, characterized by a semicircular region followed by a linear segment. This behavior confirms the electrochemical nature of the ZIHSCs system. However, the presence of an additional semicircular region suggests a more complex charge-transfer mechanism, which is further analyzed in the subsequent discussion. Notably, the ZIHSCs employing the modified SAC cathode material displayed the smallest semicircular region, indicating superior charge-

transfer properties and reaffirming earlier observations that this configuration offers higher specific capacitance or energy density. The detailed quantification of the resistance values is presented in the following discussion.

GCD analysis is another essential electrochemical technique, alongside CV and EIS, used to quantitatively evaluate the specific capacity, specific capacitance, energy density, and power density of ZIHSCs employing various cathode materials. A broader GCD curve at a constant current density indicates higher specific capacity and capacitance. Additionally, a lower slope of the GCD curve at higher potentials is associated with increased energy density.

Notably, the GCD curves of ZIHSCs with different cathode materials at a current density of 2 A/g, as shown in [Fig. 7\(d\)](#), reveal that the system utilizing the modified SAC cathode exhibits the broadest GCD profile. This indicates that it possesses the highest specific capacitance and energy density among all tested systems, supporting the findings from both CV and EIS analyses. For a more detailed quantitative evaluation, Equations [\(1\)](#), [\(2\)](#) were employed. The specific capacitance values for ZIHSCs with SC, SAC, and modified SAC cathodes were determined to be 13 F/g, 181 F/g, and 411 F/g, respectively. Correspondingly, the energy density values were 9 Wh/kg, 125 Wh/kg, and 152 Wh/kg for the same series of cathode materials.

Furthermore, as previously discussed, a comprehensive analysis of the Nyquist plot was conducted to uncover the complex mechanisms occurring within the electrodes of the ZIHSCs system in this study. Specifically, an equivalent electrical circuit model was introduced to simulate the actual processes taking place when the electrolyte interacts with the electrode material, as illustrated in [Fig. 8\(a\)](#). In this circuit model, six primary components are organized into three main regions: the bulk electrolyte region, the electrochemically inactive region, and the electrochemically active region. Zinc ions generally pass through these regions sequentially, starting with the bulk electrolyte, then moving through the electrochemically inactive region of porous carbon, and finally interacting with electrons in the electrochemically active region of porous carbon.



Subsequently, the zinc ions that reach the electrochemically active region can follow one of two mechanisms: (1) undergoing a redox reaction at the electrode surface, represented by the charge transfer constant phase element CPE_{ct} and its associated resistance R_{ct} ; or

(2) participating in Coulombic interactions with electrons at the electrode surface, characteristic of an EDLC mechanism, represented by CPE_c .

This overall process, as depicted in the equivalent circuit model, serves as a representative framework for fitting the Nyquist plots of ZIHSCs with various cathode materials. However, since the constant phase element (CPE) is frequently used in circuit models, its impedance expression must be defined. Typically, the CPE is a frequency-dependent component, and its impedance is represented by

equation (6) [26], [27], [28], [37], [50], [51], [52], [53], [54], [55], [56], [57]. (6)

In equation (6), Q represents the constant of the CPE, ω is the applied angular frequency, and n is the exponential parameter associated with the CPE. The interpretation of the CPE constant Q becomes meaningful at specific values of n : when $n = 1$, Q corresponds directly to the capacitance; when $n = 0.5$, Q is related to the diffusion coefficient of ions involved in redox reactions; and when $n = 0$, Q represents the reciprocal of resistance.

Additionally, the exponent n provides insight into the energy storage behavior of the system. Specifically, when $n = 1$, the CPE behaves like an ideal capacitor; when $n = 0.5$, it reflects Warburg-type impedance, which is characteristic of diffusion-controlled redox processes; and when $n = 0$, the CPE mimics a pure resistive element [52].

The CPE also becomes physically meaningful when placed in parallel with a resistance R . In such a configuration, the resulting RC parallel system can be approximated by an effective capacitor C_{eff} , whose capacitance is expressed as follows [58]. (7)

Equation (7) is particularly useful because it is frequency independent and can effectively represent the charge accumulated in an RC parallel system. In the context of our electrical circuit model, this approximation facilitates the interpretation of charge accumulation within the electrochemically inactive porous structure. This accumulation suggests that the inactive, porous regions impede the transport of zinc ions to the active sites on the electrode surface.

By applying equation (6) along with the principles of series-parallel connections in circuit analysis, the total impedance of the electrical circuit model (8) can be expressed as in equation (8). (8)

By employing Equation (8) and the generalized reduced gradient (GRG) nonlinear optimization method, the Nyquist plots of ZIHSCs with different cathode materials were successfully fitted. The fitting results are presented in Fig. 8(b–d), corresponding to ZIHSCs using SC, SAC, and modified SAC as cathode materials, respectively. Furthermore, the parameters of the electrical circuit model, including charge transfer resistance and those related to the inactive porous region, were successfully extracted and are presented

in [Table 1](#). Specifically, the parameter related to the inactive porous region was expressed as the effective inactive porous region capacitance ($C_{\text{eff},i}$), as defined by Equation (7).

Table 1. Extracted parameters of the equivalent electrical circuit model obtained from impedance fitting.

ZIHSCs system	$C_{\text{eff},i}$ (F/g)	R_{ct} (mΩ g)
SC//Zn	3.85	3.18
SAC//Zn	0.72	12.12
Modified SAC//Zn	0.84	4.28

Interestingly, based on the extracted parameters presented in [Table 1](#), the SC//Zn system exhibits the highest $C_{\text{eff},i}$. This suggests that ions tend to accumulate within the inactive porous structure, making it difficult for them to penetrate and reach the active sites. Although this system also shows the lowest charge transfer resistance, the limited ion accessibility to the active material diminishes the practical benefit of this low resistance. This observation aligns with the known structural characteristics of the SC cathode, which possesses a less porous architecture, thereby restricting ion interaction with electrochemically active regions.

In contrast, both SAC//Zn and modified SAC//Zn systems display lower $C_{\text{eff},i}$ values, indicating more efficient ion penetration through the inactive region. This improvement is attributed to the enhanced porosity of the SAC and modified SAC materials, which facilitates better ion transport to the active sites. However, the two systems differ in terms of charge transfer resistance: the modified SAC//Zn system demonstrates a significantly lower charge transfer resistance compared to SAC//Zn. This implies a more effective interaction between zinc ions and the electrode surface in the modified SAC system.

The reduced charge transfer resistance in the modified SAC//Zn configuration contributes to its superior electrochemical performance, including higher specific capacity and energy density. This enhancement is primarily due to the structural improvements in the modified SAC, such as increased porosity and a more ordered framework, which collectively enhance ion transport and electrical conductivity.

Alongside the detailed analysis of the Nyquist plots, an extensive CV study was conducted across a range of potential windows, from 0.9 V to 2.1 V, to determine the optimal operating voltage for the prepared ZIHSC systems. For the system using SC as the cathode material, a typical EDLC profile was observed up to 1.7 V, as shown in [Fig. S3\(a\)](#). Beyond this voltage,

a sharp increase in current response was observed, indicating the onset of gas evolution, an undesirable reaction that can compromise the stability and performance of the device.

In contrast, the ZIHSC systems utilizing SAC and modified SAC cathode materials exhibited hybrid behavior, characterized by a combination of EDLC features and redox activity with peak responses around 1.2 V, as illustrated in [Fig. S3\(b and c\)](#). However, both systems began to show signs of gas formation when the potential exceeded 2.0 V. This effect was more pronounced in the modified SAC//Zn system, likely due to the higher conductivity and more ordered carbon structure of the cathode, which enhances ion transport but may also accelerate parasitic reactions at elevated voltages. As a result, to maintain system stability and prevent degradation due to gas evolution, the maximum potential window for all subsequent experiments was limited to 2.0 V.

In addition to determining the optimal potential window for further analysis, the specific capacitance based on the CV curves (C_{sp}) was calculated for all systems across different potential windows. This calculation was performed using equation (9) [36,48].

In equation (9), i represents the current density as a function of potential, and ΔV denotes the applied potential window. Using this equation, the specific capacitance for all ZIHSC systems was successfully calculated and is graphically presented in [Fig. S3\(d-f\)](#).

Specifically, the ZIHSC system with SC as the cathode exhibited specific capacitance values of 11.3 F/g, 12.6 F/g, 13.5 F/g, 14.7 F/g, 15.4 F/g, 16.7 F/g, and 19.7 F/g at potential windows of 0.9 V, 1.1 V, 1.3 V, 1.5 V, 1.7 V, 1.9 V, and 2.1 V, respectively. In comparison, the SAC cathode system achieved higher specific capacitance values of 80.3 F/g, 101.1 F/g, 122.6 F/g, 135.3 F/g, 139.5 F/g, 141.9 F/g, and 144.7 F/g across the same range of potential windows. The modified SAC cathode system demonstrated the highest performance, with specific capacitance values of 154.1 F/g, 188.5 F/g, 227.3 F/g, 252.8 F/g, 262.9 F/g, 268.8 F/g, and 276.6 F/g, also corresponding to the same potential windows. These results further confirm the superior electrochemical performance of the modified SAC material observed in earlier analyses.

Moreover, across all tested potential windows, the specific capacitance of all ZIHSC systems increased as the potential window expanded. This trend is expected, as redox reactions generally do not occur at lower voltages. As the potential window increases and redox activity begins, the charge storage capacity, and thus the specific capacitance, also increase. For both SAC and modified SAC cathodes, the specific capacitance plateaued between 1.7 V and 1.9 V, suggesting this range as the optimal potential window for balancing performance and stability. Although a further increase to 2.1 V led to a higher apparent capacitance, this rise is likely due to gas evolution, indicated by a sharp increase

in current, rather than a true enhancement in charge storage. Therefore, limiting the operating potential window to 2.0 V is crucial for maintaining stable and reliable electrochemical performance in ZIHSCs.

The specific capacitance was also evaluated using CV measurements at different scan rates for all ZIHSC systems. In this analysis, scan rates ranging from 10 mV/s to 50 mV/s were applied. The CV curves at various scan rates for ZIHSC systems with SC, SAC, and modified SAC as cathode materials are presented in [Figure S4\(a\)](#), [Figure S4\(d\)](#), and [Figure S4\(g\)](#), respectively.

The ZIHSC system with SC as the cathode showed specific capacitance values of 22.1 F/g, 18.6 F/g, 16.5 F/g, 15.0 F/g, and 13.9 F/g at scan rates of 10 mV/s, 20 mV/s, 30 mV/s, 40 mV/s, and 50 mV/s, respectively, as illustrated in [Fig. S4\(b\)](#). In comparison, the SAC cathode system achieved specific capacitance values of 155.2 F/g, 139.9 F/g, 130.4 F/g, 125.2 F/g, and 118.4 F/g across the same range of scan rates, as shown in [Fig. S4\(e\)](#). Consistent with previous findings, the modified SAC cathode system delivered the highest performance, with specific capacitance values of 310.7 F/g, 275.9 F/g, 256.4 F/g, 242.7 F/g, and 232.3 F/g at the corresponding scan rates, as presented in [Fig. S4\(h\)](#).

In contrast to the trend observed with varying potential windows, the specific capacitance decreased as the scan rate increased. This behavior is typical in electrochemical systems; at higher scan rates, ions have less time to interact effectively with the electrode surface. As a result, there is insufficient time for ions to fully participate in the energy storage process, leading to a reduction in both specific capacity and specific capacitance in the ZIHSC systems.

Furthermore, CV at different scan rates can also be used to analyze the energy storage mechanism of the full cell system. One common approach is the power law analysis, which helps distinguish between capacitive and diffusion-controlled processes. This analysis is described by equation (10) [41,52].(10)

In equation (10), i_p represents the CV peak current, v represents the scan rate, a represents the constant, and b represents the exponential constant corresponding to the energy storage mechanism. Basically, a value of b close to 0.5 indicates more diffusive behavior and a value of b close to 1 indicates more capacitive behavior. This value of b is between these two values, indicating the hybrid mechanism.

Interestingly, power law analysis was successfully applied to all ZIHSC systems. The results are shown in [Figure S4\(c\)](#), [Figure S4\(f\)](#), and [Figure S4\(i\)](#), corresponding to ZIHSC systems with SC, SAC, and modified SAC as cathode materials, respectively. This analysis

was performed for both positive and negative scans to evaluate the reversibility of the energy storage mechanisms in the prepared ZIHSC systems.

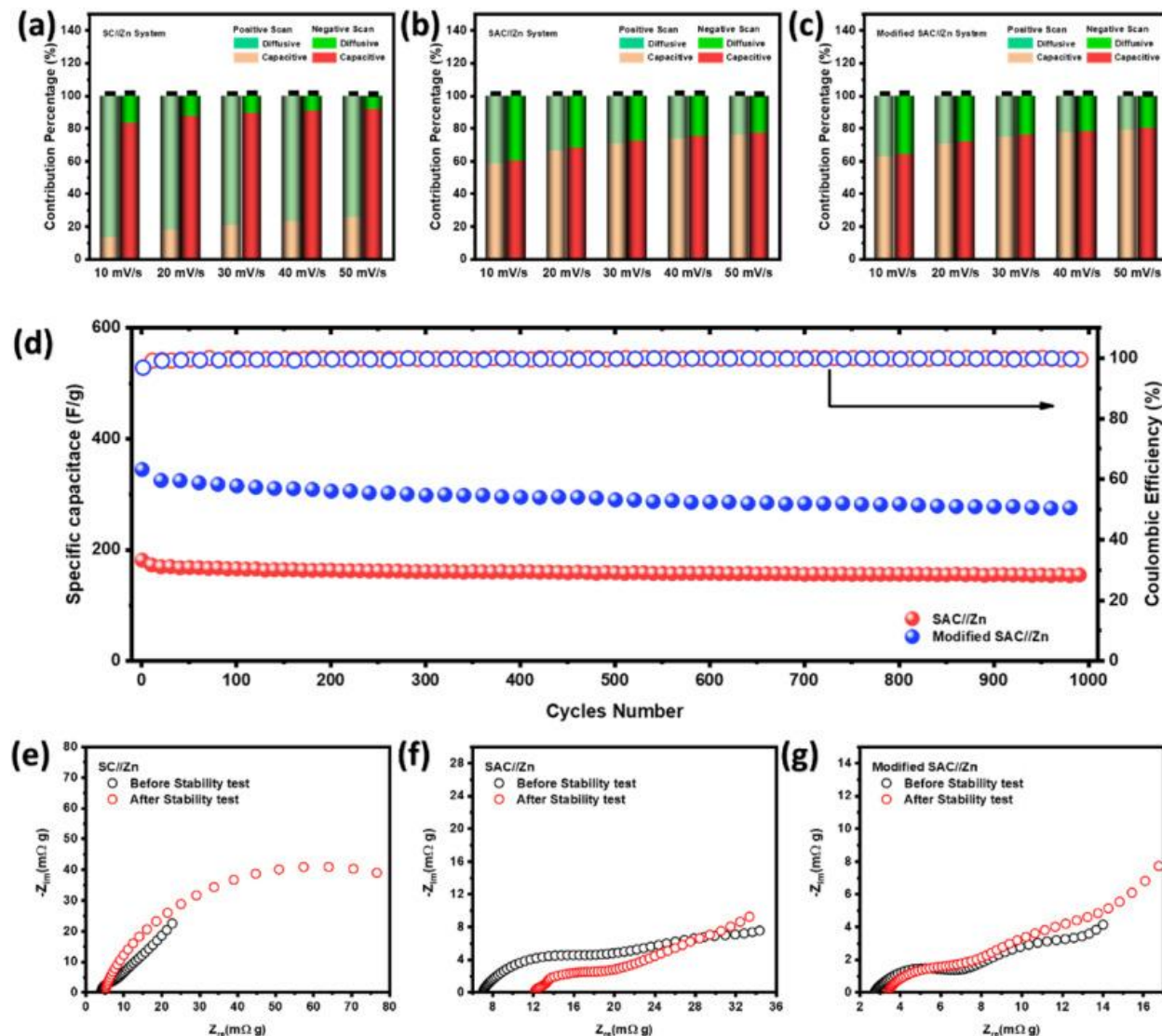
As shown in Fig. S4(c), the b -value from power law analysis of the ZIHSC system with the SC cathode reveals a notable imbalance between the positive and negative scans. In the positive scan, the b -value is close to 0.6, indicating a diffusion-controlled process, whereas in the negative scan, it approaches 1.0, which reflects ideal capacitive behavior. This inconsistency suggests an irreversible and inefficient energy storage mechanism, contributing to the lower electrochemical performance observed for this system, as previously discussed. In contrast, Fig. S4(f) illustrates that the ZIHSC system with the SAC cathode shows improved symmetry between the two scan directions, with b -values of approximately 0.84 and 0.85 for the positive and negative scans, respectively. These values indicate a hybrid mechanism with greater reversibility, suggesting that activation has enhanced the energy storage behavior. Most notably, the ZIHSC system with the modified SAC cathode, as shown in Fig. S4(i), exhibits an almost perfect balance, with both scans yielding a b -value of 0.86, which is also indicative of a reversible hybrid mechanism. This excellent symmetry supports the system's superior electrochemical performance and can be attributed to the structural enhancements in the modified SAC cathode, such as improved ion diffusion through a more open porous network and increased electrical conductivity resulting from a more ordered carbon framework.

Furthermore, in addition to the power law analysis, the Dunn method was employed as a complementary approach to determine the quantitative contribution of different energy storage mechanisms. This method relates the CV peak current to the corresponding scan rate, as described by equation (11) [27,41].(11)

In equation (11), i_p represents the CV peak current, v is the scan rate, k_1 is a constant corresponding to the capacitive mechanism contribution, and k_2 is another constant representing the contribution from the diffusive mechanism. These constants are determined by fitting the experimental data using a linearized form of equation (11). Once the values of k_1 and k_2 are obtained, the percentage contributions of the capacitive and diffusive mechanisms can be quantitatively evaluated.

Using Dunn method analysis, the percentage contributions of the energy storage mechanisms for all ZIHSC systems were successfully determined for both positive and negative scans, as shown in Fig. 9(a–c). Notably, the ZIHSC system with SC as the cathode exhibited the most unbalanced energy storage behavior between the positive and negative scans. The maximum difference in contribution between the two scan directions across all tested scan rates reached approximately 70 %, while the minimum difference was around 66 %. These large discrepancies indicate a significant imbalance in the energy storage

mechanism, which aligns with the inferior electrochemical performance previously observed for this system.



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Fig. 9. Percentage contributions of energy storage mechanisms determined using the Dunn method for ZHSC systems with (a) SC, (b) SAC, and (c) modified SAC as cathode materials. (d) Cycling stability evaluated at a current density of 4 A/g. Nyquist plots after 1000 cycles for ZHSC systems using (e) SC, (f) SAC, and (g) modified SAC as cathode materials.

In contrast, the ZHSC system utilizing modified SAC as the cathode showed the most balanced behavior. The maximum difference in energy storage contribution between the positive and negative scans was approximately 1.3 %, with a minimum difference of 0.9 %.

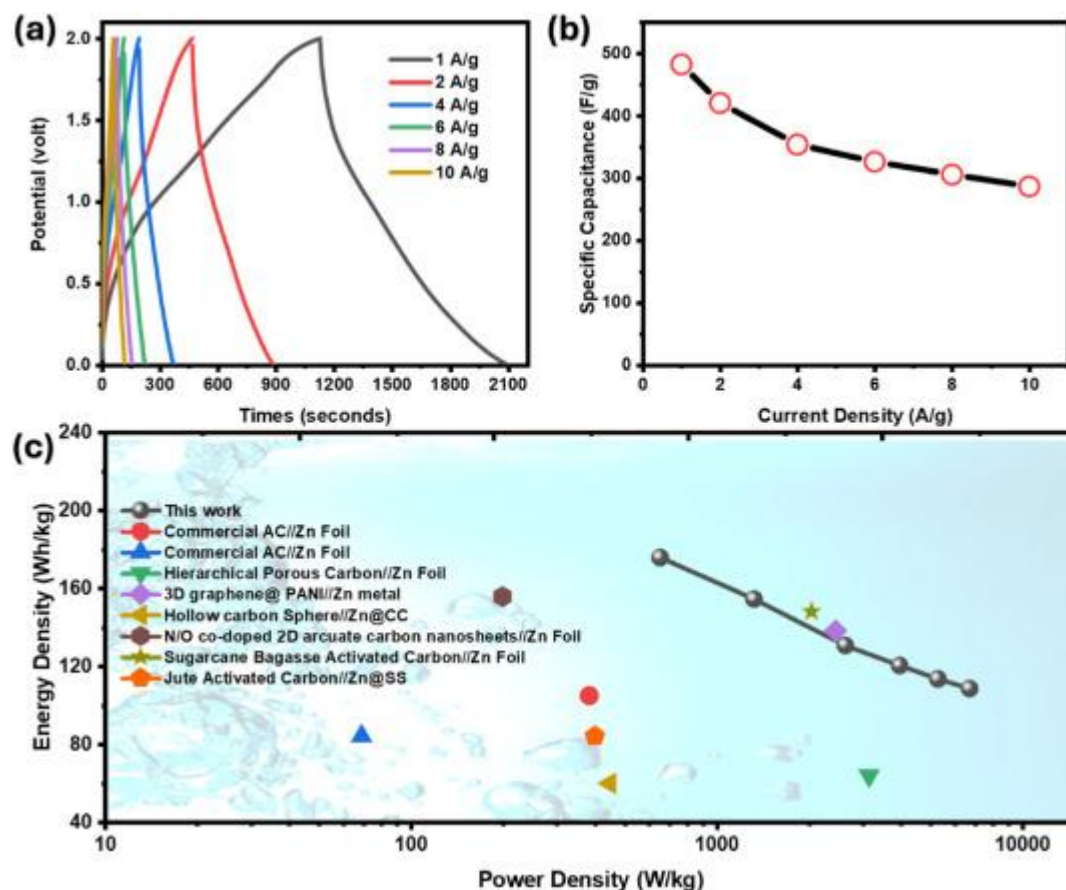
These minimal differences indicate a highly consistent and reversible energy storage process in both scan directions, further supporting the superior electrochemical performance of the modified SAC-based system. This outcome also aligns well with the findings from the earlier power law analysis.

To evaluate the long-term stability of ZIHSC systems with different cathode materials, GCD testing was performed at a moderate current density of 4 A/g over 1000 cycles. The results, presented in [Fig. 9\(d\)](#) and [Fig. S5](#), display both specific capacitance and coulombic efficiency throughout the cycling process. Interestingly, the initial specific capacitance varied significantly among the systems: SC//Zn showed a low value of 5 F/g, SAC//Zn reached 179 F/g, and the modified SAC//Zn system exhibited the highest initial capacitance at 343 F/g. This trend aligns well with earlier findings, confirming that the modified SAC cathode delivers the most enhanced electrochemical performance.

In terms of cycling stability, the modified SAC//Zn system maintained about 80 % of its initial capacitance after 1000 cycles. While slightly lower than the SAC//Zn system, which retained 85 %, this small difference is likely due to the higher electrical conductivity of the modified SAC. Greater conductivity improves charge transport but may also promote side reactions, such as gas evolution, which can lead to mild degradation over extended cycling. Nonetheless, 80 % retention is still considered excellent for high-performance energy storage systems.

Additional evidence of the modified SAC//Zn system's robustness comes from post-cycling EIS, as shown in [Fig. 9\(g\)](#). The impedance profile remained more stable compared to the SC//Zn and SAC//Zn systems shown in [Fig. 9\(e\)](#) and (f), respectively, reinforcing the conclusion that the modified SAC cathode supports a highly reversible and durable energy storage process during repeated cycling. Moreover, all ZIHSC systems maintained coulombic efficiencies near 100 %, indicating minimal charge loss and excellent reversibility during the stability test.

Furthermore, comparing only the prepared ZIHSC systems is not sufficient to provide a comprehensive understanding for the broader energy storage community. Therefore, the optimized ZIHSC system using the modified SAC cathode was selected for comparison with previously reported ZIHSC systems. To facilitate this comparison, GCD measurements were conducted at various current densities ranging from 1 A/g to 10 A/g. These measurements were used to determine the energy density and corresponding power density of the modified SAC//Zn system. The GCD curves are shown in [Fig. 10\(a\)](#), while the specific capacitance values, calculated using equation (1), are presented in [Fig. 10\(b\)](#).



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Fig. 10. (a) GCD curves of the modified SAC//Zn system at various current densities. (b) Corresponding specific capacitance values of the modified SAC//Zn system at different current densities. (c) Ragone plot comparing the performance of the ZIHSC system using the modified SAC cathode with previously reported ZIHSC systems.

Notably, the modified SAC Zn system exhibited specific capacitance values of 482 F/g (268 mAh/g), 420 F/g (233 mAh/g), 354 F/g (197 mAh/g), 326 F/g (181 mAh/g), 305 F/g (169 mAh/g), and 286 F/g (159 mAh/g) at current densities of 1 A/g, 2 A/g, 4 A/g, 6 A/g, 8 A/g, and 10 A/g, respectively. This trend of decreasing capacitance with increasing current density is consistent with the expected ion transport limitations at higher rates. It should also be noted that capacitance values obtained from CV and GCD are typically different because the two techniques probe the system under distinct electrochemical conditions. Furthermore, these capacitance values obtained by GCD measurement correspond to energy densities and power densities of 175 Wh/kg (654 W/kg), 154 Wh/kg (1.32 kW/kg), 131 Wh/kg (2.64 kW/kg), 120 Wh/kg (3.99 kW/kg), 113 Wh/kg (5.33 kW/kg), and 108 Wh/kg

(6.71 kW/kg). The resulting energy and power densities are either higher than or comparable to those reported in previously developed ZIHSC systems, as summarized in [Table 2](#).

Table 2. Comparison between the ZIHSC system using the modified SAC cathode and previously reported ZIHSC systems.

Cathode	Anode	Energy Density (Wh/kg)	Power Density (kW/kg)	Ref.
Modified SAC	Zn Foil	175	0.65	This work
Modified SAC	Zn Foil	108	6.71	This work
Commercial AC	Zn Foil	104	0.38	[59]
Commercial AC	Zn Foil	84	0.07	[60]
Hierarchical Porous Carbon (HPC)	Zn foil	64	3.16	[61]
3D graphene@ PANI composite	Zn metal	138	2.46	[62]
Hollow carbon Sphere	Zn@CC	60	0.45	[63]
N/O co-doped 2D arcuate carbon nanosheets	Zn foil	156	0.19	[31]
Sugarcane Bagasse Activated Carbon	Zn foil	148	2.05	[30]
Jute Activated Carbon	Zn@SS	84	0.40	[26]

As illustrated in [Table 2](#), the ZIHSC system integrated with the modified SAC cathode exhibits electrochemical performance that is either on par with or exceeds the performance of previously reported ZIHSC systems, particularly in terms of both energy density and power density. This observation underscores the potential of the proposed

system as a promising candidate for advanced energy storage applications. To provide a visual comparison, a Ragone plot was constructed, as presented in [Fig. 10\(c\)](#).

4. Conclusion

In conclusion, this study successfully demonstrated the fabrication of highly porous activated carbon derived from sugarcane bagasse through a templating approach using magnesium oxide (MgO) as the templating agent. The motivation for employing MgO lies in its ability to act as a removable spacer during carbonization, thereby transforming part of the microporous structure into mesopores and generating an interconnected pore network that enhances ion transport pathways. In our study, this strategy proved highly effective: the modified sugarcane activated carbon exhibited not only an enhanced porous structure but also a more ordered carbon framework, as confirmed by comprehensive material characterizations including XRD, Raman spectroscopy, BET analysis, and electron microscopy. These structural improvements proved highly beneficial for its application as a cathode material, where high ionic diffusivity and electrical conductivity are essential. As a result, the ZIHSC system using the modified activated carbon cathode achieved an outstanding specific capacitance of 482 F/g and a corresponding specific capacity of 268 mAh g⁻¹ at a current density of 1 A g⁻¹, significantly outperforming systems based on unmodified sugarcane carbon materials. Furthermore, this system exhibited excellent charge transfer efficiency and highly reversible energy storage behavior, as evidenced by cyclic voltammetry at varying scan rates and electrochemical impedance spectroscopy. A strong capacity retention of 80 % over 1000 cycles further confirms its stability and long-term performance. When compared to previously reported ZIHSC systems, the modified SAC-based ZIHSC stands out as a highly promising candidate for future energy storage applications.

CRediT authorship contribution statement

Nabila Dewi Saidhah: Writing – original draft, Visualization, Validation, Formal analysis, Data curation. **Yuda Prima Hardianto:** Writing – review & editing, Writing – original draft, Visualization, Validation, Investigation. **Abdulmajid Abdallah Mirghni:** Visualization, Validation, Formal analysis. **Ananda Sholeh Rifky Hakim:** Writing – original draft, Validation, Methodology, Data curation. **Syed Shaheen Shah:** Writing – review & editing, Visualization, Validation, Investigation. **Ade Siyanti Nurul Hidayah:** Writing – original draft, Visualization, Formal analysis. **Abduljamiu Olalekan Amao:** Writing – review & editing, Validation, Formal analysis. **Markus Diantoro:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Funding acquisition. **Md Abdul Aziz:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization.

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